

Arcadia Heroes Academy

Activity 1 - Object-Oriented Programming

This report documents a Java 17 console application that manages the heroes of the Kingdom of Arcadia. The project was organized as a Maven project and can be opened directly in IntelliJ IDEA.

1. System description

The application registers up to 20 heroes using a fixed `Hero[]` array. It supports listing registered heroes, searching by name, starting and completing missions, displaying general statistics, and producing the final kingdom report.

Three character classes are available: Warrior, Mage, and Archer. Each has its own attributes and calculates strength according to its specialization. A mission has a name, difficulty, gold reward, and lifecycle (start and completion).

2. Object-oriented concepts used

Encapsulation: fields are private and exposed through controlled methods. Abstraction: `Hero` is an abstract base class. Inheritance: `Warrior`, `Mage`, and `Archer` extend `Hero`. Polymorphism: the `Hero` array stores all subclasses and calls overridden methods such as `calculateStrength()` and `useSpecialAbility()`.

Overriding: each subclass supplies its own behavior for strength and special ability. Overloading: `Hero` provides `attack()` and `attack(Hero target)`. The program also uses methods with parameters, conditionals, repetition, primitive data types, and the required array.

3. Naming and organization

Packages and folders are in English: `model`, `service`, and `ui`. Java classes use `PascalCase`; variables, fields, and methods use `camelCase`. The source code is separated by responsibility, with console interaction isolated from business rules.

4. Execution capture

The capture below shows a validated execution scenario with one hero from each class, a completed mission, and the generated final report.

Execution sample - Java console

```
=== ARCADIA HEROES ACADEMY ===
```

```
1 - Register hero  
2 - List heroes  
5 - Start mission  
6 - Complete mission
```

```
Hero registered successfully. [Aldric - Warrior]
```

```
Hero registered successfully. [Lyra - Mage]
```

```
Hero registered successfully. [Robin - Archer]
```

```
Aldric: Mission started: Goblin Raid (Easy).
```

```
Mission completed: Goblin Raid. Reward: 100 gold.
```

```
===== KINGDOM REPORT =====
```

```
Total Heroes: 3
```

```
Warriors: 1 | Mages: 1 | Archers: 1
```

```
Average Level: 10.00
```

```
Strongest Hero: Lyra
```

```
Completed Missions: 1
```

Conclusion: the implementation meets the activity requirements and follows clean-code practices through small focused classes, meaningful English names, input validation, and separation of concerns.